

Mikel Yu

Research Interests: Scalable Oversight, Debate, Interpretability

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EDUCATION

- **Technical University of Denmark (DTU)** *Expected: Feb 2027 – Dec 2028*
Master of Science in Human-Centered Artificial Intelligence GPA: - / 12.0
- **Technical University of Denmark (DTU)** *Sep 2023 – Jun 2026*
Bachelor of Science in Artificial Intelligence and Data GPA: 9.0 / 12.0
- **Johns Hopkins University (JHU)** *Sep 2025 – Dec 2025*
Exchange Semester GPA: 3.70 / 4.00

PUBLICATIONS

- **Yu, Mikel, and Fernando, Benjamin - In review at Acta Astronautica | Preprint**  *2026*
Seismoacoustic recovery of the SpaceX CRS-32 Dragon re-entry with a proof-of-concept for automated detection
 - Designed and implemented an end-to-end signal-detection pipeline that localizes atmospheric re-entry events from open seismic network data, extending Dr. Fernando's Science-published methodology.
 - Trained a classification model to detect N-waves under significant label scarcity and uncertainty; Classifier achieved 100% recall and picks within 0.10s of manual annotation for 68% of held-out stations, despite noisy, expert-limited supervision.
- **Yu, Mikel, et al. - Manuscript in preparation**  *Feb 2026 – Jun 2026*
B.Sc. Thesis: Counterfactual Explanations for ML Triage Lyngby, Denmark
 - Developed a counterfactual explanation framework for a deployed ML-based patient acuity classification model, enabling causal what-if queries over the feature space of a live triage system.
 - Graded 12/12; both supervisor and external examiner assessed work as publication-worthy

EXPERIENCE

- **Johns Hopkins University** *Sep 2025 – Dec 2025*
Independent Undergraduate Researcher, Morton K. Blaustein Department of Earth and Planetary Sciences Maryland, USA
 - Conducted independent research under the supervision of Reines Distinguished Fellow Dr. Fernando, extending his Science-published work on atmospheric re-entry detection (see Publications).
- **Technical University of Denmark** *Jan 2025 – Jan 2026*
Teaching Assistant, Computer Science and Artificial Intelligence Lyngby, Denmark
 - Facilitated problem-solving sessions and proctored assessments for foundational Mathematics (calculus, linear algebra, differential equations) across two course iterations.
- **TopTutors** *Aug 2023 – Present*
Private Tutor, Mathematics and other high school subjects Remote
 - Ongoing private tutoring in mathematics (calculus level) for high school and middle school students.

PROJECTS

- **Adversarialness of Adversarial attacks**  *Feb 2025 – Jun 2025*
Quantification and benchmarking of adversarial attacks
 - Proposed the **true class robustness ratio (TCRR)**, a novel metric to quantify model confidence degradation in the target class under attack. Distinct from standard success-rate metrics, which only track misclassification. Benchmarked TCRR against existing robustness metrics across multiple adversarial attack methods, revealing that even when predictions remained correct, confidence in the true class often plummeted: accuracy alone is not a sufficiently descriptive metric.

AWARDS AND ACHIEVEMENTS

- **Anthropic Fellows Program — AI Safety, Anthropic, 2026:** Reached the second review round in a globally competitive fellowship.
- **Merit Award — High School Valedictorian, Hasseris Gymnasium, 2022:** Graduated with the highest GPA in the entire year, achieving a grade of 12.4/12.0 by undertaking two additional A-level subjects.

TECHNICAL SKILLS

Programming: Python, Java, R, C

Machine learning: PyTorch, TensorFlow, scikit-learn, NumPy, pandas

HPC: IBM Spectrum LSF, Linux, Git